

HD Task 4.0 - Project Proposal - Ecosave

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I. Introduction

1. Our selling point

For my project, I am proposing an application in Android that will assist users to maintain their daily budget in the process of learning about the stock market, called Ecosave. Its selling point is to be linked to a real-time S&P/ASX 200 Index API and a Llama 2 finance buddy model to provide personalized

2. Market comparison

In the current 2026 mobile market, there are many budgeting apps with an AI assistant model built into them. However, I have noticed that the models run through inference calls to online LLMs, which would compromise the user's anonymity since their spending habits would be used as training data for those LLMs.

3. The solution

By running a local Llama 2 model, Ecosave will offer individual financial analysis and it will not upload personal spending information to third-party servers.

II. Objective

The app's primary objective is to enable beginners to learn about the stock market without getting intimidated. It prevents financial literacy fatigue and provides customized budget advice as well as market overviews, all without jeopardizing the privacy or security of its user since Llama 2 will run locally.

III. Proposed Solution

1. Core features

Ecosave will have a dual interface: a budget tracker, and a dashboard showing ASX200 stock trends. Llama 2 will be directly integrated on the device using Python via Chaquopy.

2. LLM's purpose

The AI will handle four main tasks:

- ❖ Market summarization: explain ASX stock movements in simple terms
- ❖ Budget optimization: look at the user's local budget inputs to suggest personalized savings strategies
- ❖ Conversational QA: answer basic financial questions from the user
- ❖ Planning: warn the user if they might overspend based on historical data

3. LLM's benefits

Our integration of an on-device LLM guarantees data privacy and ensures that sensitive financial inputs and budgeting history is strictly contained within the user's physical device. Additionally, processing queries locally provides zero-latency responses and allows the core budgeting features to be fully accessible offline.

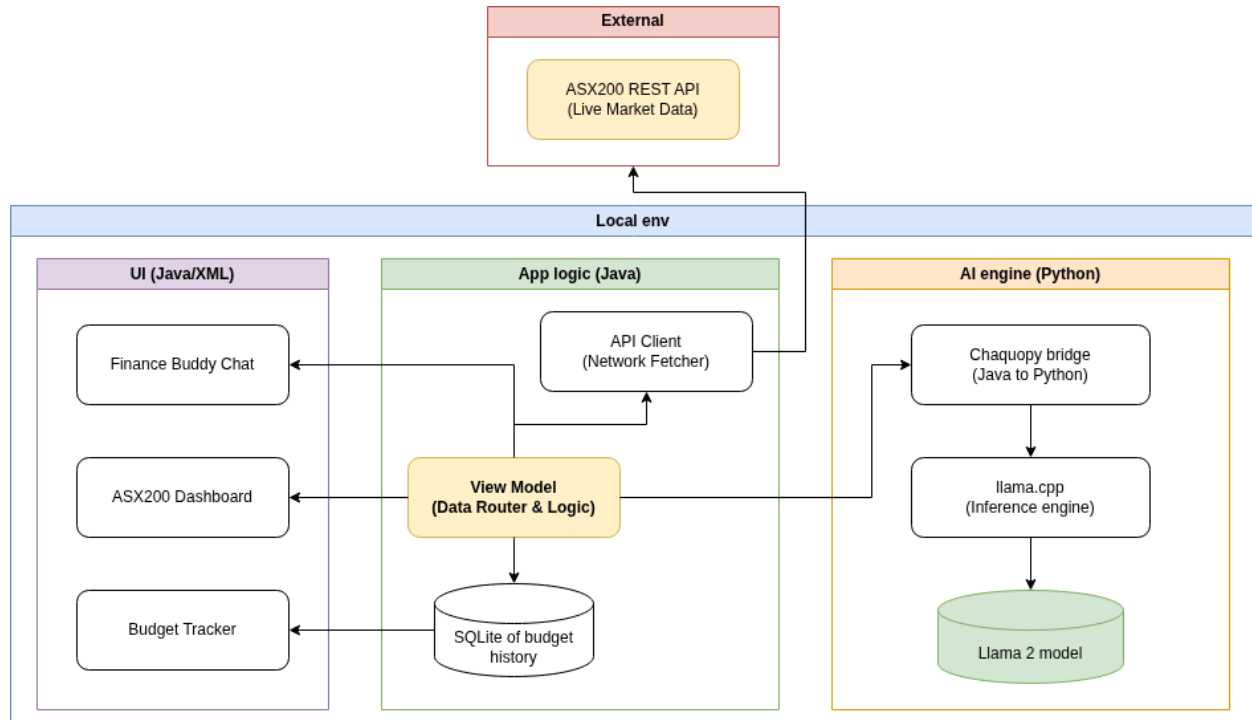
4. User safety

To guarantee safety and quality answers, I will create boundaries that are placed in the master prompt so the AI only provides financial/investment advice and is strictly focused on education. It comes with privacy as the data remains completely on the device. The Java view model will handle the data flow between the Android UI and the Python script executing Llama 2 model.

IV. Application Architecture & Tech Stack

1. Overall architecture

The app's architecture consists of one external and one on the local environment:



2. Tech stack

Here is my proposed tech stack for the application:

- ❖ Fixed stack: Android Studio, Java, Python (Chaquopy plugin)
- ❖ Inference engine: llama.cpp (llama-cpp-python) - integration with Chaquopy Python environment
- ❖ Architecture flow:
 - Android UI: display budget lists, chat interface, and ASX charts
 - View model: connects user inputs to local database & routes chat text to the AI
 - AI Engine: Llama 2 local

3. AI Model

Llama 2 was selected as the local AI for this project due to its optimal balance between performance, open-source accessibility, and local deployment viability. In addition, Meta allows free commercial and research use for the AI model. Furthermore, the model received much community support via llama.cpp, which allows it to be compressed to run efficiently on mobile hardware with limited RAM (such as the 8GB target device).

4. Deployment

The application will be configured to become deployable on a local Pixel 7 Pro emulator device on the user's machine. It will be able to run on Android Studio's emulator environment, as well as plugged in Android devices.

5. Assumptions

There are two main assumptions we need to make in order for the app to function: the testing device or emulator needs at least 8GB of RAM to run the model weights; and I can access a reliable, free REST API for the ASX200 data (Tradingview API/Finnhub API/...).

V. Development Methodology

This project will be done using an Agile development methodology. As the idea of making a large language model run at least in part on a phone is experimental, Agile enables me to develop and test features within small sprints. Also, all features, fixes, or modifications to the code will be recorded in the git tree history. By doing so, I will be able to verify the safety of my Llama 2 prompts and it will also confirm that all components of the application are working correctly, as well as receive feedback on my tutor to correct bugs.

VI. Conclusion

By successfully building and deploying Ecosave, I will demonstrate how we can integrate a local AI like Llama 2 directly on a mobile device to solve a real-world problem. I will also use my prerequisite knowledge on building personalized Agents and working with REST APIs to create a localized AI application on Android.

Links & LLM Statement

I. Github

<https://github.com/Mangon3/Ecosave>

II. LLM Declaration

No LLM was utilized during the process of completing this assessment